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BEST PRACTICE SERIES

Filter Inspection Process Using Special Filter-Cutting Bench

Maintenance and Repair Application Component Rebuild Component Rebuild Component Life Management MARC Management Com Ma

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1.0 Introduction

Oil filters should be inspected at each planned maintenance period. Typically, oil filters are cut open and inspected using a standard shop bench and workshop vice. The problem with this approach is that it is difficult to properly secure the filter in order to cut the cap off of the filter. In addition, oil contained in the filter can spill on the workbench or floor.

This Best Practice utilizes a simple, low cost filter cutting and inspection workbench as an integral part of the filter inspection process. It has a built-in retainer to securely hold the filter while cutting off the cap, an oil collection tank to contain all spills, and a shop vice to squeeze oil out of the filter paper. It can be a fixed workbench stationed in a PM area or made portable by adding wheels.

2.0 Best Practice Description

A filter inspection Best Practice uses the following procedure: (the boxed items require the filter-cutting bench and are detailed below)

1. During a standard service (or after 50 hours after an engine change), the technician will remove the filters from the engine and place them in a drain tray for the oil to drain from them.
2. The oil filter is then placed in the filter-cutting bench and the top is removed from the filter.
3. A section of paper is cut from the filter.
4. Oil from the section of paper is squeezed from the filter.
5. The filter paper is inspected for abnormal material.
6. If no abnormal material is found, the filter is discarded.
7. If abnormal material is found, another section of filter paper is cut to obtain more material for inspection.
8. The technician, Workshop Fitter and Workshop Foreman should attempt to identify the source of the material. The filter paper is bagged and tagged for safekeeping.
    a. If the abnormal material is identified and is not of a serious nature, the machine will be sent back to work and another filter will be cut after 50 hours for monitoring purposes.
    b. If the abnormal material can be identified and it is of a serious nature, then inspection of the engine will start. A typical example: brass material from the pump drive thrust washer.

If the abnormal material cannot be identified, then a sample should be sent to SOS for a Filtergram. The results of the Filtergram will determine further action.

The filter-cutting bench provides a safe, simple, and consistent way to inspect filters. Basic bench design includes a cylinder with clamp screws as a filter holding fixture, a waste oil collection tank, and a mounted vise. Modifications can be made to suit individual needs, including wheels, alternate filter securing methods, or a smaller press for removing oil from the filter paper. The bench can be placed in the PM bay so technicians can immediately cut and inspect filters while the machine is still in the shop for the PM.

Basic directions for using the bench:

1. Install the filter and tighten the setscrews. Remove the filter cap with the filter-cutting tool.

2. Place the filter on the bench in preparation for cutting out a section of paper to be inspected.

3. Remove a section of filter paper from the filter.

Note: Users may wish to install a mechanism for holding the filter while cutting the paper.

4. Place the filter paper in the vice and remove the oil from the filter paper. Inspect the filter for abnormal material.

3.0 Implementation Steps

1. Must have and follow a standard practice of filter inspections as part of a condition monitoring process.
2. Personnel must be trained on the filter inspection process and interpretation of findings.
3. Design and fabricate test bench to meet needs.
4. Add, “Filter cut and inspected” to the PM checklist and sign off.
5. Ensure adequate lighting and magnifying glasses are available for proper filter paper inspection.

4.0 Benefits

- Provides a safe and secure method of holding the filter while cutting it.
- Captures all oil released from the filter - prevents oil spills and environmental issues.
- All oil from the filter cutting process is captured in an oil storage tank.
- Cutting bench serves as a physical reminder for technicians to cut filters, ease of use promotes good practice.
- The cutting of filters becomes an integral part of condition monitoring.

5.0 Resources Required

- Fabrication costs are approximately US $620 per bench.

6.0 Supporting Attachments / References

Filter Inspection Bench Details.pdf

7.0 Related Best Practices

None.

8.0 Acknowledgements

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